

f:

```
In [2]: from sympy import *
t, R = symbols("t, R")
```

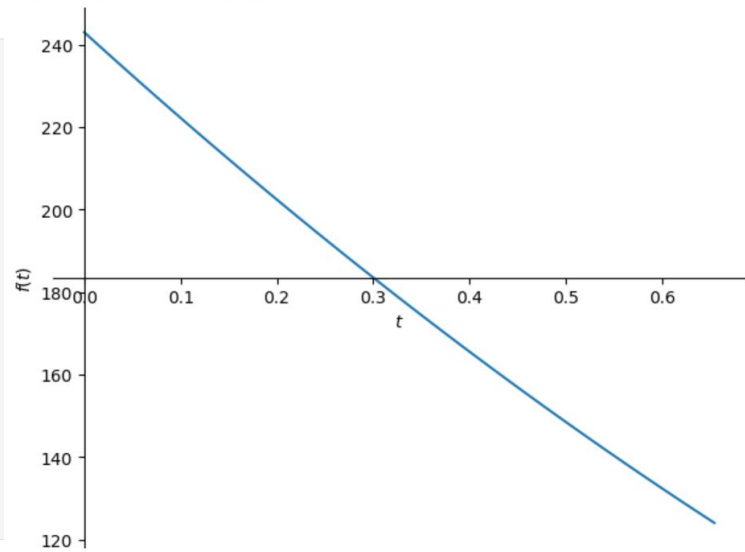
```
m = 6.0 #kg
g = 9.82 #m/s^2
μk = 0.4
v0 = 9.0 #m/s
```

```
v = v0 - μk*g*t
ω = 5/2*μk*g/R*t
I = 2/5*m*R**2
t1 = 2*v0/(7*μk*g)
```

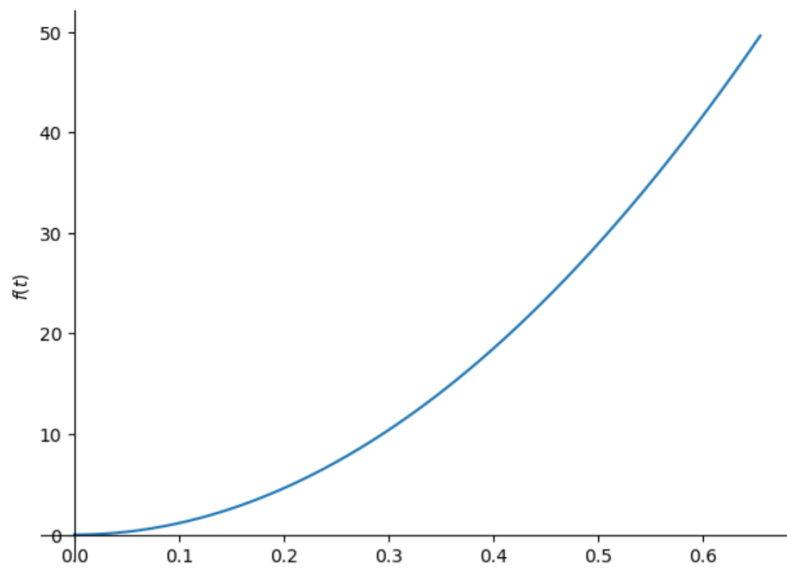
```
K_trans = 1/2*m*v**2
K_rot = 1/2*I*ω**2
K_tot = K_trans + K_rot
```

```
plot(K_trans, (t, 0, t1))
plot(K_rot, (t, 0, t1))
plot(K_tot, (t, 0, t1))
```

Translatoriske kinetiske energi:



Rotationelle kinetiske energi:



Totale kinetiske energi:

